Roll No. | | | | | | | |

## MID SEMESTER EXAMINATION, SEPTEMBER 2025

Name of the Course: M. Tech Biomedical Engineering
Name of the Paper: Biomedical Instrumentation

Semester: 1
Paper Code: BEM 102

Time: 90 Minutes
Marks: 50

Maximum

**Note:**

(i) Answer all the questions by choosing ONE of the sub questions.
(ii) Each question carries 10 marks.

| | | |
|---|---|---|
| **Q1** | (10 marks) | **CO1** |
| (a) | "A man instrument system is a combination of a human subject and electronic instruments used to measure physiological or other human-related phenomena safely and accurately." Justify | |
| | OR | |
| (b) | Measuring a living system is very challenging because it encounters several problems. Throw some light on these problems. | |
| **Q2** | (10 marks) | **CO2 & CO6** |
| (a) | Define the four physiological systems of the human body. Explain comprehensively the Cardiovascular System | |
| | OR | |
| (b) | Elucidate the Respiratory system in detail | |
| **Q3** | (10 marks) | **CO2 & CO3** |
| (a) | What are Bioelectric potentials? Give a comprehensive analysis of an ECG. | |
| | OR | |
| (b) | Why do we use Electroretinogram and Electrooculogram? Explain the process of measuring both of these. | |
| **Q4** | (10 marks) | **CO2** |
| (a) | What are transducers? Describe the transduction principle with a valid example. | |
| | OR | |
| (b) | What are active and passive transducers? Differentiate between them. | |
| **Q5** | (10 marks) | **CO2** |
| (a) | Explain the resting and action potentials. Also discuss the importance of Na pump. | |
| | OR | |
| (b) | Explain Electroencephalogram in details. Include in your answer all the ranges of EEG frequencies. | |